

Chapter 1

Methodology and Results Interpretation

This chapter explains the empirical strategy used to study how monetary conditions are associated with foreign direct investment (FDI) inflows in the country-year panel. The central question is whether monetary liquidity and interest-rate proxies move in directions that are consistent with the expected relationship between domestic monetary conditions and FDI inflows. The evidence is deliberately interpreted as conditional panel association rather than causal identification. Across the preferred fixed-effects specifications, broad money generally has the expected positive sign, while deposit, real, and lending-rate proxies generally have the expected negative sign. Most headline estimates are statistically weak. The lending-rate coefficient is negative and statistically significant, but it is treated as robustness evidence because lending-rate coverage is structurally sparse.

1.1 Data Sources and Sample Construction

The analysis uses a harmonized country-year panel that combines FDI inflows, monetary aggregates, interest-rate indicators, and macroeconomic controls for ASEAN-oriented economies. FDI, broad money, trade openness, inflation, GDP per capita, population, tourism, and human-capital variables are drawn from cleaned international macroeconomic data. Real and lending interest rates are prioritized from the World Development Indicators to ensure consistent definitions across countries. The model set is organized *ex ante* around theoretically motivated monetary-policy channels: liquidity, deposit-rate, real-rate, and lending-rate channels.

The final analytical panel covers 2010–2023 and contains 154 country-year observations. It is unbalanced because several monetary and structural variables are not observed for every country-year. The countries in the sample are Brunei Darussalam, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Singapore, Thailand, Timor-Leste, and Viet Nam. The ASEAN framing is therefore regional, with Timor-Leste included because it is part of the available country-year sample.

1.2 Data Processing and Variables

The dependent variable is FDI net inflows as a percentage of GDP. The main monetary variables are broad money as a percentage of GDP, the deposit interest rate, the real interest rate, and the lending interest rate. The baseline controls are inflation measured by the GDP deflator, trade as a percentage of GDP, log GDP per capita, and exchange-rate depreciation. The exchange-rate depreciation variable is computed as

$$xr_dep_it = 100 [\ln(exchange_rate_{it}) - \ln(exchange_rate_{i,t-1})],$$

where the exchange rate is the local-currency-per-U.S.-dollar series. Positive values therefore indicate local-currency depreciation. Winsorized versions of FDI and exchange-rate depreciation are used only for descriptive and visual sensitivity checks; the reported regressions use the observed variables in the main specification.

Missingness is handled according to the variable role. The dependent variable and key monetary proxies are observed-only: country-years without the required proxy are not imputed into the regression sample. Eligible controls are interpolated or edge-filled only where the preprocessing rules permit this treatment. Structural gaps are preserved and reported. This distinction matters because missing monetary-proxy coverage changes the realized estimation support across models, especially for real and lending interest rates.

1.3 Empirical Specification

The preferred empirical specification is a two-way fixed-effects panel model:

$$FDI_{it} = \beta_1 Monetary_{it} + \gamma' \mathbf{X}_{it} + \alpha_i + \lambda_t + \varepsilon_{it},$$

where FDI_{it} is FDI inflows as a percentage of GDP for country i in year t , $Monetary_{it}$ is the monetary-policy proxy selected for the model, $\mathbf{X}_{it}$ is the vector of controls, α_i denotes country fixed effects, and λ_t denotes year fixed effects. The estimator reported in the main tables is fixed effects with Driscoll–Kraay standard errors using a Bartlett kernel covariance estimator. This covariance choice is appropriate because the diagnostic tests frequently reject homoskedasticity, while the panel is short and potentially exposed to common regional shocks.

Pooled OLS, random effects, Hausman diagnostics, variance inflation factors (VIF), Breusch–Pagan and White tests, and Pesaran CD tests are used as diagnostics and comparators. They are not treated as the primary thesis estimator. The estimates are consequently interpreted as within-country conditional associations after absorbing country and year effects.

1.4 Descriptive Statistics

Table 1.1 reports descriptive statistics for the final analytical sample. FDI inflows are highly dispersed: the mean is 5.424 percent of GDP, the standard deviation is 6.825, and the observed range runs from -16.205 to 32.596 percent of GDP. The upper tail is partly driven by Singapore, where high FDI-to-GDP observations are economically plausible for an entrepot financial and investment hub.

These observations are retained because they reflect meaningful country characteristics rather than obvious measurement errors.

Monetary variables differ substantially in coverage. Broad money is observed in 133 country-years, deposit interest rates in 132, and real and lending interest rates in 117 each. Exchange-rate depreciation also has substantial dispersion, with a mean of 5.137 percent, a standard deviation of 40.232, and a maximum of 476.795 percent. This high maximum is retained as an observed value but reinforces the need to interpret exchange-rate coefficients cautiously.

Table 1.1: Descriptive statistics for the processed panel

Variable	Count	Mean	Std. dev.	Min.	Median	Max.	Missing (%)
FDI (% GDP)	154	5.424	6.825	-16.205	3.501	32.596	0.0
Broad money (% GDP)	133	83.757	38.329	23.572	79.496	148.651	13.6
Deposit interest rate (%)	132	2.960	3.012	0.122	1.577	13.994	14.3
Real interest rate (%)	117	5.095	8.451	-30.123	4.594	41.294	24.0
Lending interest rate (%)	117	8.411	4.418	3.060	5.680	22.612	24.0
Inflation, GDP deflator (%)	154	4.153	8.302	-20.822	3.071	59.080	0.0
Trade (% GDP)	140	126.336	80.037	32.972	116.238	379.099	9.1
Log GDP per capita	154	8.460	1.263	6.701	8.115	11.411	0.0
Exchange-rate depreciation (%)	143	5.137	40.232	-9.431	0.543	476.795	7.1

Notes: The regression dependent variable is FDI net inflows as a percentage of GDP. This table is descriptive rather than estimated; the regression tables below use two-way fixed effects with Driscoll–Kraay standard errors. Significance stars are not applicable. Counts differ because the analytical panel is unbalanced and variables have different observed coverage.

1.5 Model Roles and Sample Coverage

The empirical design separates the analysis into headline models and robustness or appendix models. M1 is the main liquidity-channel model. M2 is the main deposit-rate-channel model. M3 is a lagged deposit-rate robustness model. M4 uses the real interest rate as the rate proxy and is retained as a main real-rate-channel specification. M5 uses the lending rate and is reported as robustness only because the lending-rate series has narrower structural coverage. M6 and M7 add tourism and human-capital controls and are treated as appendix sensitivity checks rather than headline evidence.

Table 1.2 summarizes the fixed-effects Driscoll–Kraay sample support for the reported models. The full analytical panel has 154 observations, but complete-case estimation leaves 112 observations in M1, 96 in M2, 91 in M3, 96 in M4, and 96 in M5. Lao PDR is a structural gap for broad money and deposit-rate specifications, with only one non-missing proxy year. Cambodia has no observed real-rate or lending-rate proxy years, and Lao PDR again has only one observed year for those proxies. The real-rate and lending-rate models therefore do not estimate the same country-year support as the broader liquidity model.

1.6 Main Regression Results

Table 1.3 presents the main fixed-effects results. M1 estimates a positive broad-money coefficient of 0.0670 with a Driscoll–Kraay standard error of 0.0578. This sign is consistent with the expected liquidity-channel relationship, but the estimate is not statistically significant. M2 again estimates a positive broad-money coefficient, 0.0732, and a negative deposit-rate coefficient, -0.2940 , but both

Table 1.2: Model roles and realized FE-DK sample sizes

Model	Thesis role	Observations	Countries	Structural monetary-proxy coverage note
M1	Main liquidity channel	112	9	Lao PDR: 1/14 broad-money years
M2	Main deposit-rate channel	96	8	Lao PDR: 1/14 deposit-rate years
M3	Lagged deposit-rate robustness	91	8	Lao PDR: 1/14 deposit-rate years
M4	Main real-rate channel	96	8	Cambodia: 0/14; Lao PDR: 1/14 real-rate years
M5	Lending-rate robustness	96	8	Cambodia: 0/14; Lao PDR: 1/14 lending-rate years
M6a	Appendix tourism sensitivity from M2	96	8	Lao PDR: 1/14 deposit-rate years
M6b	Appendix tourism sensitivity from M4	96	8	Cambodia: 0/14; Lao PDR: 1/14 real-rate years
M7a	Appendix human-capital sensitivity from M2	83	7	Lao PDR: 1/14 deposit-rate years
M7b	Appendix human-capital sensitivity from M4	83	7	Cambodia: 0/14; Lao PDR: 1/14 real-rate years

Notes: The regression dependent variable is FDI net inflows as a percentage of GDP. FE-DK denotes two-way fixed effects with Driscoll–Kraay standard errors using Bartlett kernel covariance. Significance stars are not applicable in this sample-summary table. Sample sizes vary because the panel is unbalanced and each model requires complete observations for its selected monetary proxy and controls.

coefficients are imprecise. The M2 trade coefficient is -0.0381 and statistically significant at the 10 percent level, which is not aligned with the expected positive trade-openness sign.

M4 gives the same broad qualitative message for the real-rate channel. Broad money is positive at 0.0986 , while the real interest rate is negative at -0.0911 , but neither estimate is statistically significant. M3 provides weak lagged-proxy evidence: the lagged deposit-rate coefficient is -0.0241 with a large standard error of 0.4672 . M5 is the only model in the main table with a statistically strong monetary-policy proxy. The lending-rate coefficient is -0.9684 , significant at the 1 percent level, and is consistent with a cost-of-capital interpretation. However, M5 is not promoted to the headline conclusion because lending-rate coverage is structurally sparse and its diagnostics show stronger cross-sectional-dependence concerns.

1.7 Diagnostics and Robustness Interpretation

The diagnostic results in Table 1.4 support the cautious interpretation used in this chapter. White test p -values are small in the main models, reinforcing the decision to report heteroskedasticity- and dependence-robust Driscoll–Kraay standard errors rather than relying on conventional pooled standard errors. Pesaran CD p -values are not statistically significant for M1 through M4, but M5 has a p -value of 0.0152 . This means the model with the strongest lending-rate coefficient is also the one with the clearest cross-sectional-dependence diagnostic warning among the main columns.

The fit statistics are modest. Within- R^2 values are close to zero or negative for the main headline specifications, indicating limited within-country explanatory power after absorbing country and year effects. This does not by itself invalidate the coefficient estimates, but it does limit the strength of any substantive claim. The chapter therefore emphasizes sign consistency, statistical precision, sample support, and robustness status rather than predictive fit.

Table 1.3: Main fixed-effects Driscoll–Kraay regression results

	M1 Liquidity	M2 Deposit rate	M3 Lagged deposit	M4 Real rate	M5 Lending rate
Broad money (% GDP)	0.0670 (0.0578)	0.0732 (0.0786)		0.0986 (0.0662)	0.0497 (0.0605)
Deposit interest rate (%)		-0.2940 (0.3102)			
Deposit interest rate, lag 1 (%)			-0.0241 (0.4672)		
Real interest rate (%)				-0.0911 (0.0977)	
Lending interest rate (%)					-0.9684*** (0.3467)
Trade (% GDP)	-0.0311 (0.0213)	-0.0381* (0.0224)		-0.0379 (0.0261)	-0.0547** (0.0221)
Inflation, GDP deflator (%)	0.0244 (0.0910)	0.0357 (0.0936)			0.0323 (0.0788)
Log GDP per capita	-3.1347 (3.6327)	-3.5097 (3.8685)		-3.4257 (3.7277)	-4.8413 (3.9939)
Exchange-rate depreciation (%)	0.0130 (0.0773)	0.0508 (0.1012)		0.0147 (0.0726)	0.1147 (0.0992)
Observations	112	96	91	96	96
Countries used	9	8	8	8	8
Within R^2	-0.0114	-0.0376	0.0168	0.0150	0.0021
Slope-adjusted within R^2	-0.0591	-0.1076	-0.0534	-0.0398	-0.0651

Notes: The dependent variable is FDI net inflows as a percentage of GDP. All columns use two-way fixed effects with country and year effects and Driscoll–Kraay standard errors using Bartlett kernel covariance. Standard errors are in parentheses. *, **, and *** denote statistical significance at the 10, 5, and 1 percent levels. The panel is unbalanced, and sample sizes vary across columns because monetary-proxy and control coverage differs by model.

Table 1.4: Compact diagnostics for the reported models

Model	Within R^2	Adjusted R^2	Max VIF	White p	Pesaran CD p
M1	-0.011	-0.184	11.11	0.000	0.460
M2	-0.038	-0.226	11.20	0.000	0.277
M3	0.017	-0.276	11.41	0.034	0.803
M4	0.015	-0.167	10.71	0.000	0.345
M5	0.002	-0.070	22.63	0.000	0.015
M6a	-0.178	-0.219	201.80	0.002	0.463
M6b	-0.113	-0.161	95.14	0.000	0.551
M7a	0.113	-0.167	387.19	0.062	0.112
M7b	0.117	-0.148	329.29	0.004	0.107

Notes: The dependent variable in the corresponding regressions is FDI net inflows as a percentage of GDP. Reported fit statistics correspond to two-way fixed-effects specifications with Driscoll–Kraay standard errors in the regression tables. Significance stars are not applicable in this diagnostic table. Diagnostic samples vary because the panel is unbalanced and each model uses a different complete-case support.

1.8 Economic Interpretation

The empirical pattern is directionally plausible but statistically fragile. The positive broad-money coefficients in M1, M2, M4, and M5 are consistent with a liquidity or financial-depth channel in which deeper monetary and financial conditions are associated with higher FDI inflows. However, none of the broad-money estimates is statistically significant in the reported FE-DK table. The evidence therefore suggests a possible liquidity association but does not establish a precisely estimated relationship.

The interest-rate results are also mixed. The negative deposit-rate, lagged deposit-rate, and real-rate coefficients are consistent with the view that higher domestic financing costs may be associated with lower FDI inflows, but these estimates are not statistically significant. The lending-rate coefficient is negative and statistically significant, which is the clearest cost-of-capital signal in the analysis. Still, because Cambodia has zero non-missing lending-rate years and Lao PDR has only one, this result should be read as suggestive robustness evidence from a restricted sample rather than as the chapter's primary result.

The control coefficients further caution against a strong structural interpretation. Trade openness is negative in the reported models and statistically significant in M2 and M5, contrary to the expected positive sign. Log GDP per capita is negative and imprecise in the reported contemporaneous models. These signs may reflect within-country changes over a short period, reduced estimation samples, or correlation among macro controls rather than stable cross-country structural relationships. The most defensible conclusion is that monetary conditions appear relevant to FDI inflows in directionally plausible ways, but the short unbalanced panel and proxy-specific coverage gaps make the estimates too imprecise for strong causal claims.

1.9 Limitations

Several limitations shape the interpretation. First, the panel is short, with only 14 annual observations per country before missingness. Two-way fixed effects absorb persistent country differences and common yearly shocks, leaving limited within-country variation for the monetary variables. Second, the monetary proxies are not equally observed. Real and lending interest rates have 37 missing observations each, and their structural gaps change country support. Third, macroeconomic controls such as trade openness, GDP per capita, and exchange-rate depreciation may respond to the same shocks that influence FDI. The fixed-effects design controls for time-invariant country heterogeneity and common year shocks, but it does not provide an exogenous monetary-policy shock or a full causal identification strategy.

1.10 Appendix: Additional Diagnostics and Sample Audits

Additional appendix diagnostics include correlation matrices, pairwise sample counts, VIF results, sample-loss audits, heteroskedasticity diagnostics, Pesaran CD diagnostics, and missingness summaries. These materials document the sample restrictions behind the main table and provide the basis for extended robustness discussion.

The sample audit shows that moving from M1 to M2 reduces the realized sample from 112 to

96 rows and from nine to eight countries. The main recorded loss driver is deposit-rate coverage, with the country loss note identifying Cambodia and Philippines. Moving from M2 to the lagged M3 specification reduces the sample from 96 to 91 observations, driven by lagged exchange-rate depreciation availability. Adding the human-capital control in M7 reduces the sample to 83 observations and drops Timor-Leste from the realized sample. These audits explain why M6 and M7 are kept as sensitivity branches rather than promoted to headline evidence.

1.11 Appendix: Tourism and Human-Capital Branches

The M6 tourism branches and M7 human-capital branches test whether the main monetary relationships are sensitive to additional structural controls. They are useful for appendix discussion but should not replace the headline models. M6a and M6b add log tourism arrivals to the M2 and M4 branches. M7a and M7b add the human-capital index to the M2 and M4 branches. These additions are conceptually relevant but narrow the interpretation because they increase collinearity risk, preserve the monetary-proxy coverage gaps, and, in the human-capital branches, reduce the realized sample to 83 observations.

1.12 Appendix: Review Flags and Outliers

Reviewed outliers are retained rather than mechanically deleted. These include extreme inflation observations for Timor-Leste in 2021 and Viet Nam in 2010, high FDI-to-GDP observations for Singapore, and high trade-openness observations for Singapore. These values are economically interpretable in the relevant country contexts and are retained in the main analysis. Their presence nevertheless supports the chapter’s restrained language and the use of fixed effects, robustness checks, and appendix diagnostics.